

CodeTutor - Intelligent Programming Tutoring System

An LLM-powered intelligent tutoring system for programming education

In Dialogue Tutor mode, I can answer your programming questions, explain concepts, and help you understand code. I won't give direct answers but guide you to think.



What is a variable in Python?



Great question! Understanding variables is fundamental to programming in Python. Let's break down what a variable is, how you use them in Python, and provide some practical examples.

What is a Variable?

A **variable** in Python (and many other programming languages) is a name given to a storage location which holds data values. These names act as references to the actual data stored in memory. Variables allow you to store and manipulate data within your program.

Key Points about Variables:

1. **Dynamic Typing:** In Python, you don't need to declare the type of a variable explicitly. The interpreter automatically infers the type based on the value assigned.
2. **Mutable vs Immutable:** Some types like lists and dictionaries are mutable (can be changed after creation), while others like integers and strings are immutable (cannot be changed once created).
3. **Naming Conventions:** Variable names can contain letters, numbers, and underscores but cannot start with a number. They are case-sensitive.

How to Define Variables in Python

Here's how you define and assign values to variables:

```
# Assign an integer value
age = 25

# Assign a string value
name = "Alice"

# Assign a floating-point value
height = 5.9

# Assign a list (mutable)
items = [1, 2, 3]

# Assign a tuple (immutable)
coordinates = (0, 0)

# Assign a dictionary (mutable)
person = {"name": "Bob", "age": 30}
```

Practical Examples

Let's look at each example more closely:

1. Integer Assignment

```
age = 25
print(age) # Output: 25
```